

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: ITA 06
LEARNING

COURSE NAME: MACHINE

COURSE OUTCOME COVERED

CO2: *Neural Network Representation, Perceptron Model, Multilayer Perceptrons, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)*

Assessment Tool Weightage: 40%

QUESTIONS: 15

Total Marks:25

Annexure A

Analytical Problem Solving

Code of Conduct:

*I Sarshini.S – 192321113 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:
SARSHINI

Annexure A

Short Answer Test

1, A simple perceptron model is used to classify whether a student passes or fails based on two input features, namely study hours and attendance. The weights associated with these inputs are 0.6 and 0.4 respectively, and the bias value is -0.5 . For a given student with study hours equal to 2 and attendance equal to 1, compute the net input to the perceptron and apply the step activation function to determine the final output. Based on the result, conclude whether the student passes or fails

Given:

- Weights:
 $w_1 = 0.6$ (study hours)
 $w_2 = 0.4$ (attendance)
- Inputs:
 $x_1 = 2$ (study hours)
 $x_2 = 1$ (attendance)
- Bias: $b = -0.5$

Step 1: Compute Net Input

The perceptron net input is:

$$\begin{aligned}\text{Net Input} &= (w_1 \cdot x_1) + (w_2 \cdot x_2) + b \\ &= (0.6 \times 2) + (0.4 \times 1) + (-0.5) \\ &= 1.2 + 0.4 - 0.5 = 1.1\end{aligned}$$

Step 2: Apply Step Activation Function

Step function rule:

- If net input $\geq 0 \rightarrow$ Output = 1 (Pass)
- If net input $< 0 \rightarrow$ Output = 0 (Fail)

Since:

$$1.1 \geq 0$$

Output = 1

Final Result:

- **Net Input = 1.1**
- **Output = 1 → PASS**

Conclusion:

The student **passes** based on the perceptron model.

2. In a neural network, a single neuron is trained using the backpropagation learning rule. The initial weight is 0.5 and the learning rate is 0.1. For an input value of 2, the actual output produced by the neuron is 0.6, whereas the target output is 1. Calculate the error in the output and determine the weight update using the delta rule. Hence, compute the new updated weight after one iteration of training.

Given:

- Initial weight $w = 0.5$
- Learning rate $\eta = 0.1$
- Input $x = 2$
- Actual output $y = 0.6$
- Target output $t = 1$

Step 1: Calculate Error

$$\text{Error} = t - y = 1 - 0.6 = 0.4$$

Step 2: Apply Delta Rule (Weight Update)

Delta rule formula:

$$\begin{aligned}\Delta w &= \eta \times (t - y) \times x \\ \Delta w &= 0.1 \times 0.4 \times 2 = 0.08\end{aligned}$$

Step 3: Compute New Weight

$$\begin{aligned}w_{\text{new}} &= w + \Delta w \\ w_{\text{new}} &= 0.5 + 0.08 = 0.58\end{aligned}$$

Final Result:

- **Error = 0.4**
- **Weight Update = 0.08**
- **New Weight = 0.58**

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	25	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand